

Exercise 2. E-commerce Platform Search Function

Scenario:

1. Big O notation is a way to estimate how much time an algorithm may take when the amount of data increases. It helps us understand whether an algorithm will remain efficient when the number of products on the platform becomes very large. By using Big O notation, different algorithms can be compared and the most suitable one can be selected.

Search Cases:

- **Best Case:** The required product is found immediately.
- **Average Case:** The product is found after checking some of the available products.
- **Worst Case:** The product is found at the end or does not exist in the collection.

2. Product Class

```
class Product{
    int productID;
    String productName;
    String category;

    public Product(int productID, String productName, String category){
        this.productID = productID;
        this.productName = productName;
        this.category = category;
    }
}
```

Linear Search

```
public Product linearSearch(Product[] products, String name){
    for(Product product : products){
        if(product.productName.equals(name)){
            return product;
        }
    }

    return null;
}
```

Binary Search

```
public Product binarySearch(Product[] products, String name){
    int low = 0;
    int high = products.length - 1;
    while(low <= high){
        int mid = low + (high - low) / 2;
        int compare = products[mid].productName.compareTo(name);
        if(compare == 0){
            return products[mid];
        }
        if(compare < 0){
            low = mid + 1;
        }
        else{
            high = mid - 1;
        }
    }
    return null;
}
```

4. Analysis

Linear Search

In Linear Search, products are checked one by one until the desired product is found. If the product is located near the beginning, the search finishes quickly. However, if it is located near the end, many comparisons are required.

- Best Case: $O(1)$
- Average Case: $O(n)$
- Worst Case: $O(n)$

Binary Search

Binary Search works by repeatedly dividing the sorted product list into two parts and eliminating half of the remaining elements after each comparison. Because of this approach, the number of comparisons is greatly reduced.

- Best Case: $O(1)$
- Average Case: $O(\log n)$
- Worst Case: $O(\log n)$

For a large e-commerce platform, Binary Search is a more practical choice because the product catalog can contain thousands of items. Instead of checking products one after another, Binary Search quickly narrows down the search area, resulting in faster response times. The only requirement is that the products must be arranged in sorted order before performing the search. Therefore, for large-scale applications where search operations are performed frequently, Binary Search offers better efficiency than Linear Search.